

Automatic Extraction of PDF Table Data Based on Deep Learning

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Abstract

This article delves into the automatic extraction technology of PDF table data based on deep learning, with a particular focus on the application and performance of the PaddleOCR framework in this field. With the advent of the information age, PDF documents have become one of the primary formats for data exchange and information storage, and table data, as an important component thereof, renders its automatic extraction technology particularly crucial. Subsequently, the article highlights the key features and applications of the PaddleOCR framework in extracting PDF table data. As a powerful OCR tool, PaddleOCR supports multi-task learning, including text detection, text recognition, and layout analysis, among others, enabling it to comprehensively understand the structure of PDF documents, especially those containing tables. The article conducts a detailed evaluation of PaddleOCR's performance in extracting PDF table data. By comparing the recognition results with the original tables, it is found that PaddleOCR performs exceptionally well when processing most tables, with a recognition accuracy rate of up to 93%. However, when dealing with complex tables containing merged cells and watermarks, issues such as redundant characters and recognition errors still exist. In response to these issues, the article proposes optimization solutions, such as utilizing the PP-PicoDet model for layout analysis and incorporating a watermark removal step, ultimately increasing the recognition accuracy rate to 96.4%. Finally, the article summarizes the advantages and disadvantages of PaddleOCR in extracting PDF table data and proposes future research directions.

CCS Concepts

• **Computing methodologies** → Artificial intelligence; Natural language processing; Information extraction.

Keywords

Table extraction, Deep learning, Optical character recognition

ACM Reference Format:

Wen Yang and Min Yu. 2025. Automatic Extraction of PDF Table Data Based on Deep Learning. In *2025 4th International Conference on Big Data*,

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BDICN 2025, Guangzhou, China

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ACM ISBN 979-8-4007-1242-5/2025/01

<https://doi.org/10.1145/3727353.3727391>

Information and Computer Network (BDICN 2025), January 10–12, 2025, Guangzhou, China. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3727353.3727391>

1 Introduction

Tang Haojin provided a detailed introduction to the technical difficulties in table recognition, including table borderline recognition, table restoration based on border lines, handling of complex first-row and first-column tables, merging of cross-page tables, and table data formatting. The effectiveness and performance of the system were verified by testing and evaluating the table data of the three major financial statements in PDF financial reports [1]. Tian Cuihua, Zhang Yiping, Hu Zhigang, et al. utilized Python's built-in libraries and adopted specific algorithms to quickly and accurately recognize the text information of different structured tables in PDFs. The restored table structures and text information were imported into Word and Excel documents, and the recognition accuracy was superior to tools such as Swift PDF Converter and Smallpdf [2]. Shang Yingmei proposed a new universal table extraction tool that combined data mining technology and machine learning methods and achieved an F1 score of 0.9156 on the ICDAR 2013 table dataset [3]. Li Zhoujun, Jin Daisheng, Xiao Wukui, et al. extracted table features, located and parsed the internal structure, divided the table into a two-dimensional grid structure, making it possible to obtain the position and size of each cell, and finally output the extraction results in HTML format [4]. Sebastian Schreiber proposed a new deep learning-based method for table structure recognition, which could identify the positions of rows, columns, and cells in the detected tables and demonstrate good recognition results [5]. Saman Arif and Faisal Shafait showed the performance improvement of table detection technology. They adopted the deep learning-based Faster R-CNN to detect table regions from document images [6]. Rahul Anand, Hye-Young Paik, and Cheng Wang proposed a deep learning-based method to achieve SQL-like query and analysis of financial tables in PDF annual reports [7]. Zheng Xinyi, Doug Burdick, Lucian Popa, Xu Zhong, and Nancy Xin Ru Wang propose the Global Table Extractor (GTE), a framework for a visual-guided system for connection table detection and cell structure recognition. It can be built on any object detection model [8]. E. Stricker and M. E. Scheurer developed the PDF Data Extractor (PDE) package. The tool allows users to categorize large numbers of full-text articles in PDF format, export articles containing tables as Excel tables (pdf2table), and extract relevant data using a simple user interface without requiring bioinformatics skills [9].

The above review describes the form recognition technology. Looking at various review articles, the main problems faced by form recognition, solution ideas, and tool implementation are summarized. It can be seen from a number of papers that the current research on table recognition algorithms is developing towards a diversified and high level, and it also shows the development trend of table recognition, that is, more and more scholars begin to use deep learning algorithms for table recognition research. However, what is slightly lacking is the failure to present the details of the research on the specific use of a certain technology.

In this paper, PaddleOCR, a deep learning OCR framework, is selected as the focus to discuss the leading advantages and good effects of deep learning OCR technology applied to PDF table data extraction. PaddleOCR is the OCR(Optical character recognition) function based on PaddlePaddle deep learning platform. As an open-source project, PaddleOCR provides a collection of OCR solutions, including text detection, recognition, and table information, and structured information in images. The highlight of PaddleOCR in table information extraction is the processing of PDF table data.

1.1 Key Features of PaddleOCR

Multi-task support. PaddleOCR is not only for word processing, it can carry out multi-task learning, including text detection, text recognition, layout analysis, and other functions, which will help PaddleOCR to reflect fuller thinking when processing PDF documents with rich table content. Not only can you get the text information of the document, but also through the layout analysis, you can get more content about the layout, so as to get the data of the table.

In order to obtain efficient multi-task adaptability, PaddleOCR adopted an end-to-end deep learning model, which can automatically learn and summarize important features in documents from large-scale training samples, adapt to table extraction tasks of various documents, and have extremely high end-to-end learning efficiency, greatly strengthening the generalization ability of the model to problems in actual use.

In addition, PaddleOCR supports multiple languages, making it easier for users to use when conducting research and applications across borders. Whether it is English, Chinese, or any other language, PaddleOCR can accurately read the data extracted from the table by relying on its own text detection and text recognition. PaddleOCR supports multiple languages as a selling point and has a wider range of uses and advantages in the international environment.

1.2 PaddleOCR application in PDF table data extraction

Table detection: PaddleOCR, in the form OCR, the text detection part is optimized, can quickly detect the form area in the PDF document, even a very simple single-row single-column table, or across multiple pages of multiple rows and columns of complex tables can be quickly located, providing a good basis for the following structured data extraction.

The layout analysis module is the key module for PaddleOCR to complete the table structure analysis. In this module, we can obtain the structure information such as row and column information and

header position in the table, so as to understand the content of the table and its distribution and correlation, facilitate the understanding and analysis of the table structure, and then simplify the data analysis and processing after the table structure analysis.

In the process of table content extraction, the text recognition part of PaddleOCR also performs well. The recognition part can recognize various contents in the table, including numbers, letters, and Chinese characters, and extract them accurately, which is the advantage brought by the advanced attention mechanism and sequence model of PaddleOCR, which can maintain good performance even in the changeable table environment and meet the needs of users. The table analysis process is shown in Figure 1.

2 Extraction Process

2.1 Converting PDF to Images

First, PDF preprocessing is a very important link in data processing or text processing. The PDF is split into images for each page to facilitate subsequent analysis. The use of PyMuPDF for this task is an efficient Python library.

PyMuPDF is actually an efficient Python wrapper around the MuPDF tool. MuPDF is a small but complete set of PDF and XPS and ebook viewer/renderer/tool, maintained and developed by Artifex Software, which supports PDF, of course also supports XPS, OpenXPS, CBZ, EPUB, MOBI, FB2 ebooks, etc., with extremely high performance and excellent rendering quality. Be praised and recognized by the industry.

In the processing of PDF using PDF parser, it can be learned that in general, PDF pages are composed of some blocks with borders, and these blocks are arranged in the order of their respective positions. PyMuPDF will generate Textpage object, and PyMuPDF also provides the `extractDICT()` and `extractRAWdict()` methods, it is easy to find all text and graphical information on the page, including its content, location, attributes, etc., to provide support for the following analysis work.

2.2 Layout Analysis

First, the image is analyzed using Layout-Parser. In the layout analysis process, the areas in the image are divided into five categories: text, headings, images, lists, and tables. For the first four types of regions, PP-OCR is used to complete the text detection and recognition of the corresponding regions. For the table area, after the table structuring process, the table image is converted into an Excel file with the same table style.

PP - OCR (PaddlePaddleOpticalCharacterRecognition) text detection model using MobileNetV3 as the backbone, complete the direction detection and classification. Compared with other models in the same prediction time, this model achieves higher accuracy. In order to balance accuracy and efficiency in direction classification, MobileNetV3_small_x0.35 is selected as the backbone. In order to make the model performance higher, the introduction of two kinds of image data enhancement algorithm, namely basic data enhancement algorithm (BDA) and RandAugment algorithm, the image to be used for training classifier. The setting of image resolution has a direct influence on the recognition accuracy in the model. PP-OCR sets the normalized image height and width to 48,192 respectively to increase the accuracy of the direction classifier. On the quantitative

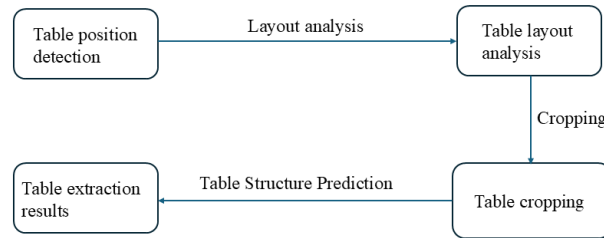


Figure 1: Flow chart

model in order to reduce the volume, and accelerate the calculation, the PACT (PARAMeterizedClippingactivation) quantitative method. However, due to the activation functions of MobileNetV3 being ReLU and hard swish, general PACT quantization results in high losses. Therefore, the PACT quantization method, which adds some extra computational steps, is used in the system and combined with some design decisions given in this paper, which is an important condition for the system to obtain the text recognition task performance of the PP-OCR text detector model.

2.3 Table Recognition

After the layout analysis is completed, table recognition is the key step to convert the table image recognition into an Excel document, which not only includes the detection and recognition of the internal text of the table but also includes the structure prediction and cell coordinate prediction of the whole table. Table recognition includes three models which are also its core content, and different models undertake different tasks:

- The single-line text detection—The DB model is responsible for accurately locating the position of each line of text in the form image, and outputting the coordinate information of each line of text, which provides the basis for the subsequent text recognition.
- Single-line text recognition—The CRNN model, based on the text coordinates detected by the DB model, recognizes the text in the table and converts image-based text information into editable text data.
- Table Structure and Cell Coordinate Prediction—The RARE model focuses on analyzing the overall structure of a table, predicting the four-point coordinates of each Excel cell and the row and column information of a table, and providing key data for table reconstruction. The specific table reconstruction flow chart is shown in Figure 2 below:

The difficulty of table recognition lies in two aspects, which not only tests the depth of technology, but also challenges the intelligence and flexibility of the algorithm.

First, the structure of the table is extracted accurately. As a form and storage of important data, the presentation of a table is complicated. Some tables have regular rows and clear cell borders, while others have irregular rows or unclear cell borders because of the design, layout, and quality of the scan. All these greatly test the difficulty of extracting the structure of the table. In order to accurately extract the structure of the table, it is necessary to accurately extract the table's rows, columns, and cell borders. This puts forward high requirements for the accuracy, stability, and robustness of the table structure extraction algorithm.

The second challenge is the integration of table information and OCR(Optical character recognition) information. OCR is able to recognize the text in the table, but how to accurately map this text to the corresponding cell becomes another important challenge. In the actual use process, the scanning quality, text font, text structuring, and other aspects will lead to some errors in the OCR-recognized text, therefore, it is necessary to introduce a mechanism to integrate the OCR-recognized text and the table structure, that is, integrate the corresponding relationship between the text and the cell to complete the integration of the OCR table.

The table recognition process is divided into two stages. The first stage is the common OCR process, which is the premise of table recognition, including two stages: text detection and text recognition. Text detection is used to locate the text area in the table and extract the text coordinate information. Text recognition converts the detected text area into editable text content. Through these two stages, the text information and coordinate information in the table can be obtained, which lays the data foundation for the prediction and processing of the table structure in the following paper.

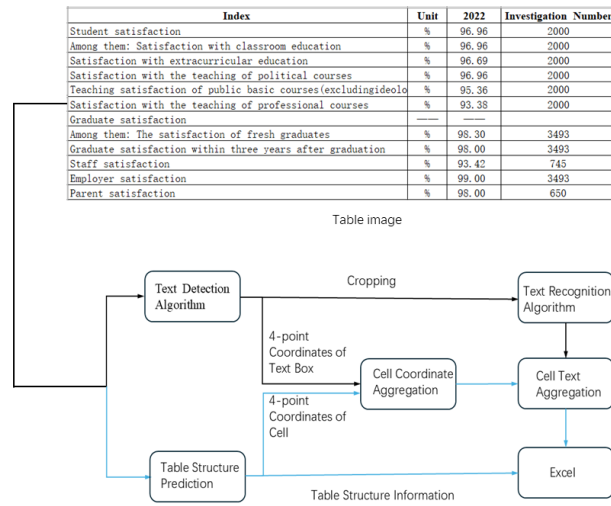


Figure 2: Table recognition flow chart.

Step 2: Predict and process the table structure. This module obtains cell coordinates and structure information. Based on coordinate information in text detection, it uses a cell coordinate aggregation module and cell text aggregation module to aggregate and connect texts belonging to the same cell, and uses an Excel export module to obtain standard Excel table data in combination with table structure information. Completed the conversion from table image to Excel table.

3 Result Analysis

Our total of 10 PDF datasets is the main research basis for our table recognition task. Through preprocessing, we found that the parts to be extracted were mainly distributed in the last 6 pages of each PDF, with a total of 60 pages of PDF to be processed. These target pages not only contain more information but also have their own characteristics, which put forward higher requirements for the identification framework we designed.

A common feature of this kind of target web page is that there are a large number of merged cells, which makes the structure of the table more complicated, and the web page is covered by a watermark, which not only affects the visual observation but also poses a greater challenge to automatic recognition. Therefore, it is particularly important to do further feature research on this kind of target web page in order to design a recognition framework more accurately and efficiently.

After we completed the layout analysis and table recognition, we compared the recognition results with the original table and found that the recognition accuracy rate was 93%, which is a very good achievement. During the recognition, we can capture every element in the table, but there are some miscellaneous characters in the recognition of elements, and the recognition error of some merged cells is relatively large. This requires the identification results to be calibrated in the later stage.

First, the layout analysis algorithm adopted in the experiment is based on PP-YOLOv2, a high-performance detection algorithm under PaddleDetection. PP-YOLOv2 is an object detection model that solves the problems in other models well, but it is not perfect. Its disadvantage may be that the positioning is not accurate, which is related to the accuracy of the algorithm to the image. At the same time, PP-YOLOv2 may have a relatively low recall rate compared to adopting the regional proposal approach, which is found in the YOLO series. In addition, the cross-phase strategy in PP-YOLOv2 reduces the use of explicit feature mapping for replication for connections, and there are still some performance barriers.

Finally, in order to further improve the speed and accuracy of layout analysis and processing, in this work, based on the lightweight layout analysis model PP-PicoDet, we customized the image scale in this scene and adopted the FGD knowledge distillation algorithm to optimize it. Finally, the layout analysis was completed at the speed of CPU41ms (excluding data preprocessing time). About 50ms).

In terms of the selection of a table recognition algorithm, we use TableRec-RARE, an end-to-end table recognition algorithm designed based on RARE, the text recognition algorithm of our initial experiment. The output result is the HTML representation of the table structure, which can easily output Excel files to meet practical requirements. In order to improve performance, we use SLANet (StructureLocationAlignmentNetwork, structure position alignment network) to increase the understanding of the model table structure.

In addition, it was noted that the watermark of the original file would interfere with table recognition. After adding the watermark removal process to the program, the sample was tested again, and the accuracy rate was increased from 93% to 96.4%, which verified the correctness of the improved direction.

4 Conclusion

Although PaddleOCR supports the recognition of tables in PDF documents well, the resolution of cells in tables with complex structures (combining cells across columns and rows, etc.) is difficult to implement automatically. Optimize the model architecture and loss function, further improve the model's ability to analyze the complex table structure, and combine cell analysis and cell recognition.

For documents in different fields and formats, the generalization performance of the model can be improved by increasing the diversity of training samples and using transfer learning and other techniques.

In this way, we can study more explicable deep learning models and develop interpretation technologies, which can not only enhance users' understanding of the model parsing table process to a certain extent, but also make users more familiar with the process, thus greatly improving users' efficiency, but also increase users' trust in the extraction results.

Acknowledgments

GrantSponsor:School-level Research Project of Shandong Yingcai University; GrantNumber:YCKY23002.

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